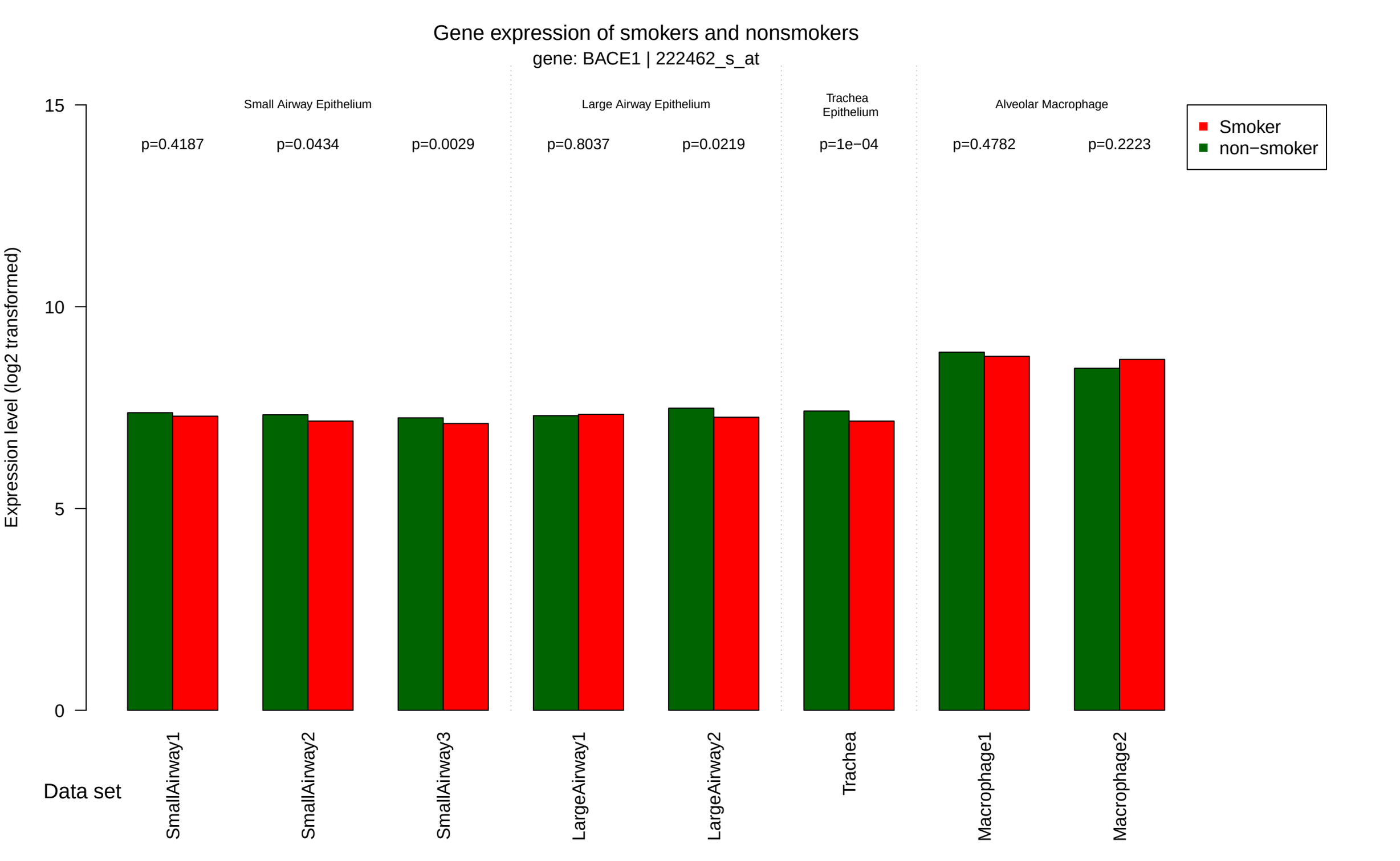
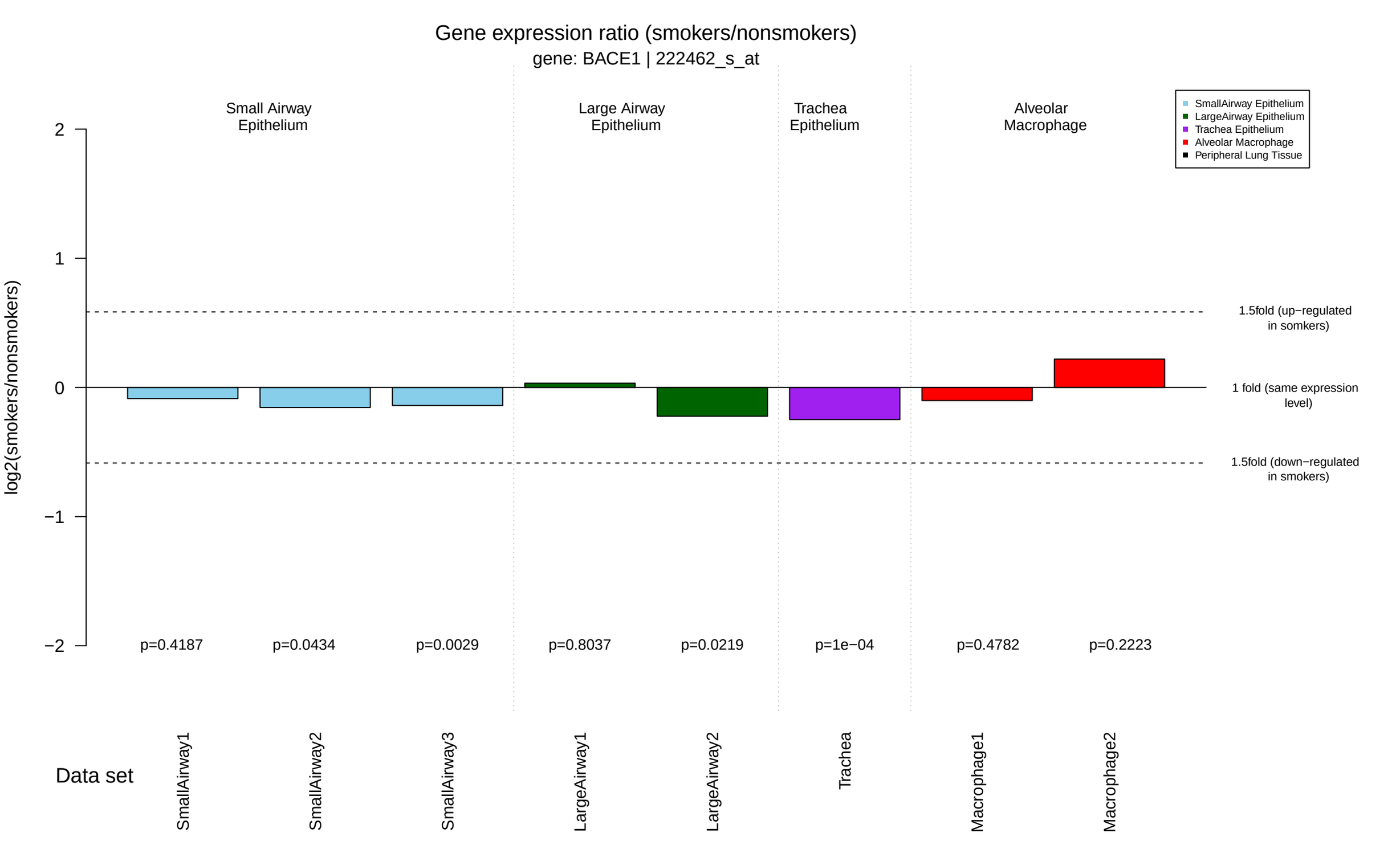
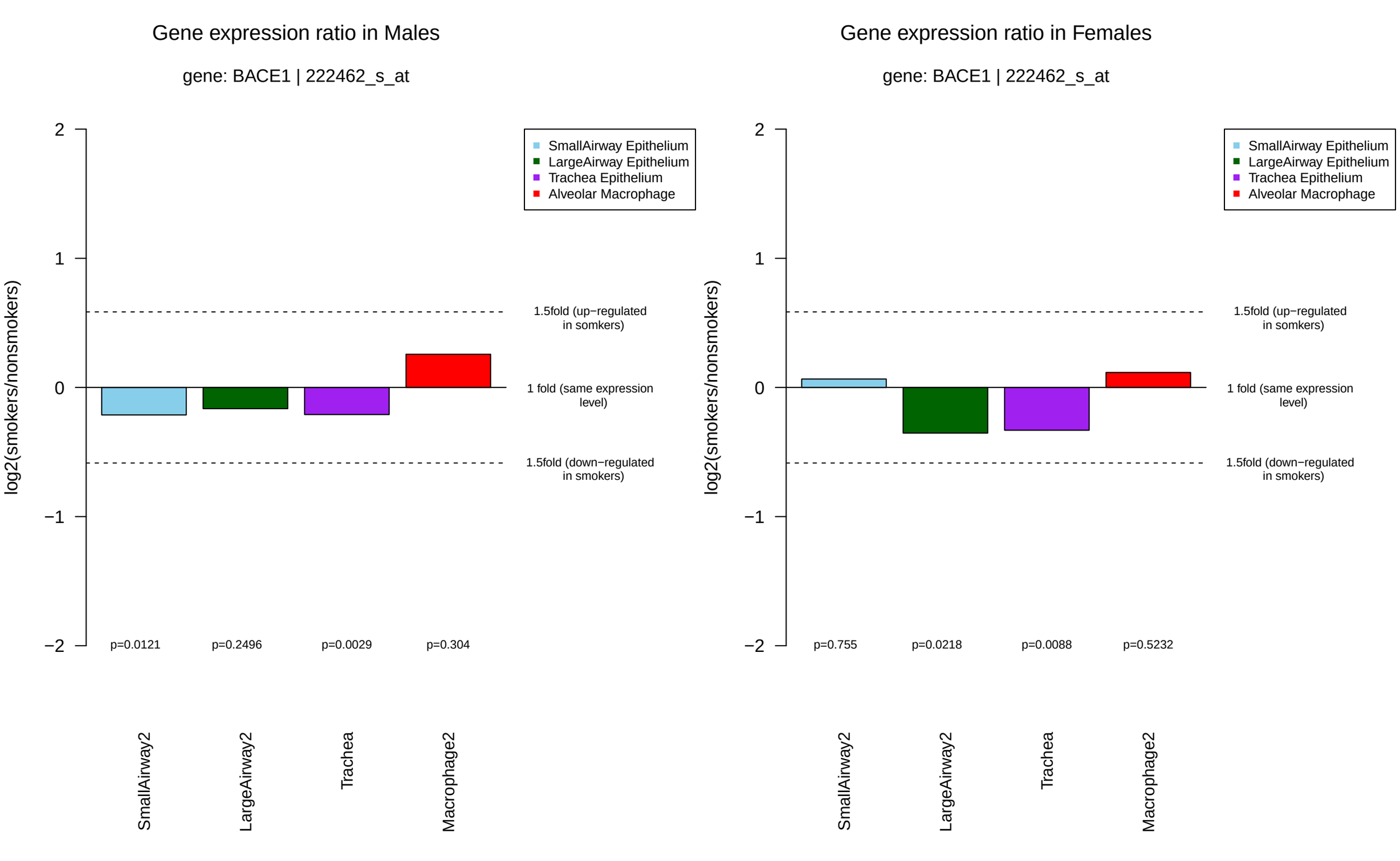
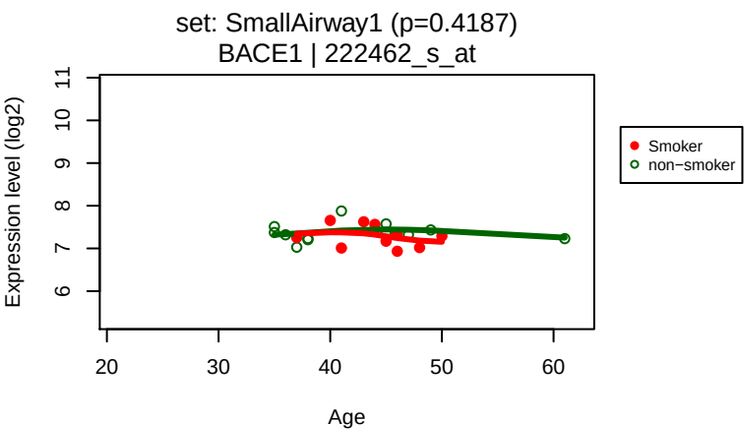
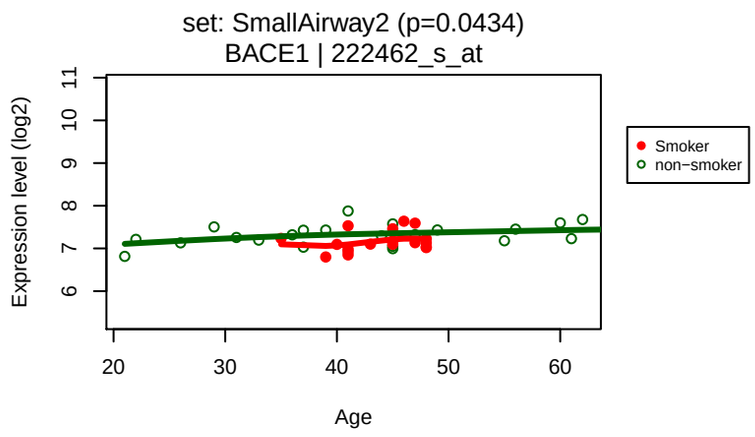
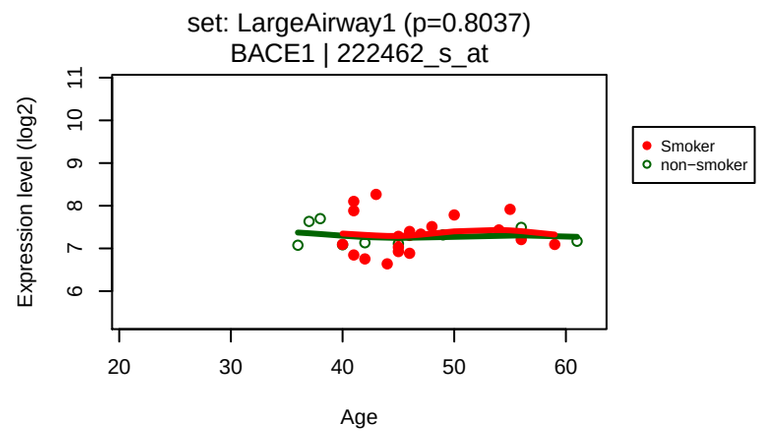

Gene expression of smokers and nonsmokers
gene: BACE1 | 222462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4187
p=0.0434
p=0.0029
p=0.8037
p=0.0219
p=1e−04
p=0.4782
p=0.2223
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

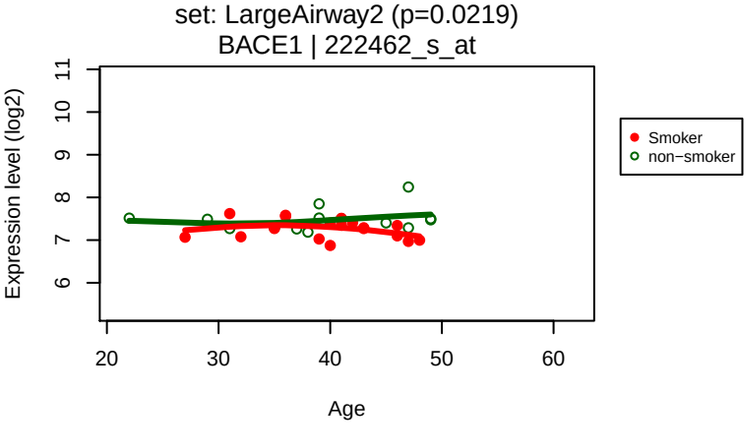

Gene expression ratio (smokers/nonsmokers)
gene: BACE1 | 222462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4187
p=0.0434
p=0.0029
p=0.8037
p=0.0219
p=1e−04
p=0.4782
p=0.2223
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

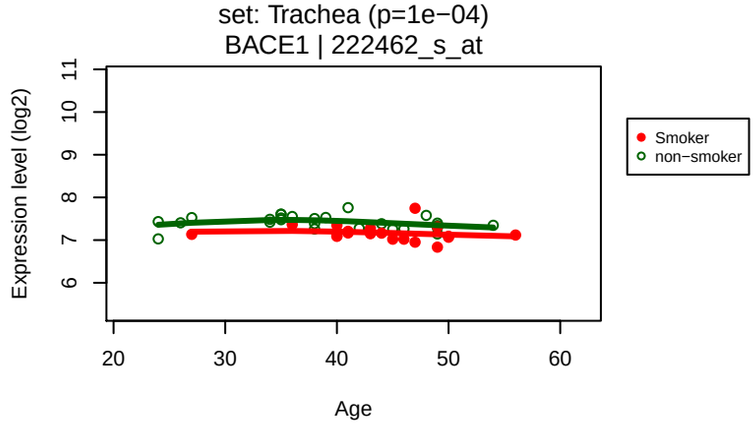

Gene expression ratio in Males
gene: BACE1 | 222462_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0121
p=0.2496
p=0.0029
p=0.304
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BACE1 | 222462_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.755
p=0.0218
p=0.0088
p=0.5232
SmallAirway2
LargeAirway2
Trachea
Macrophage2

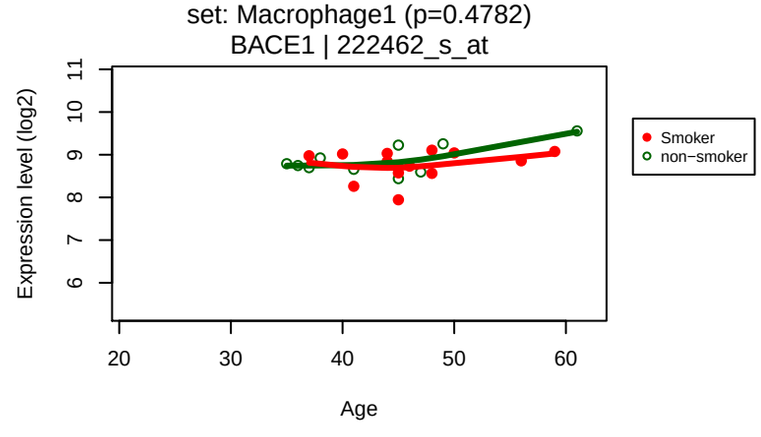

set: SmallAirway1 (p=0.4187)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker

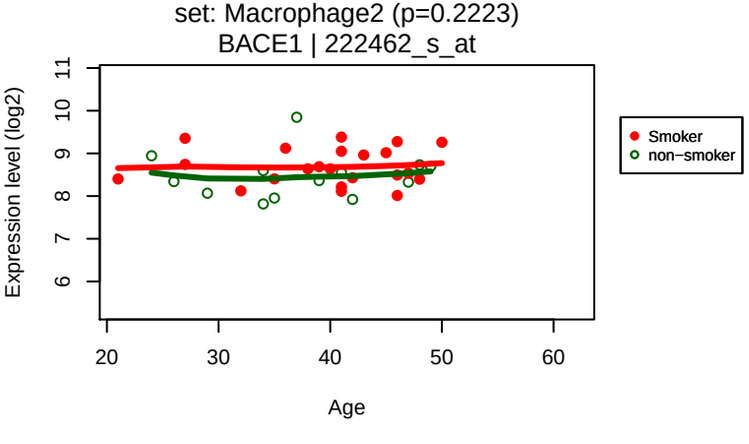

set: SmallAirway2 (p=0.0434)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker

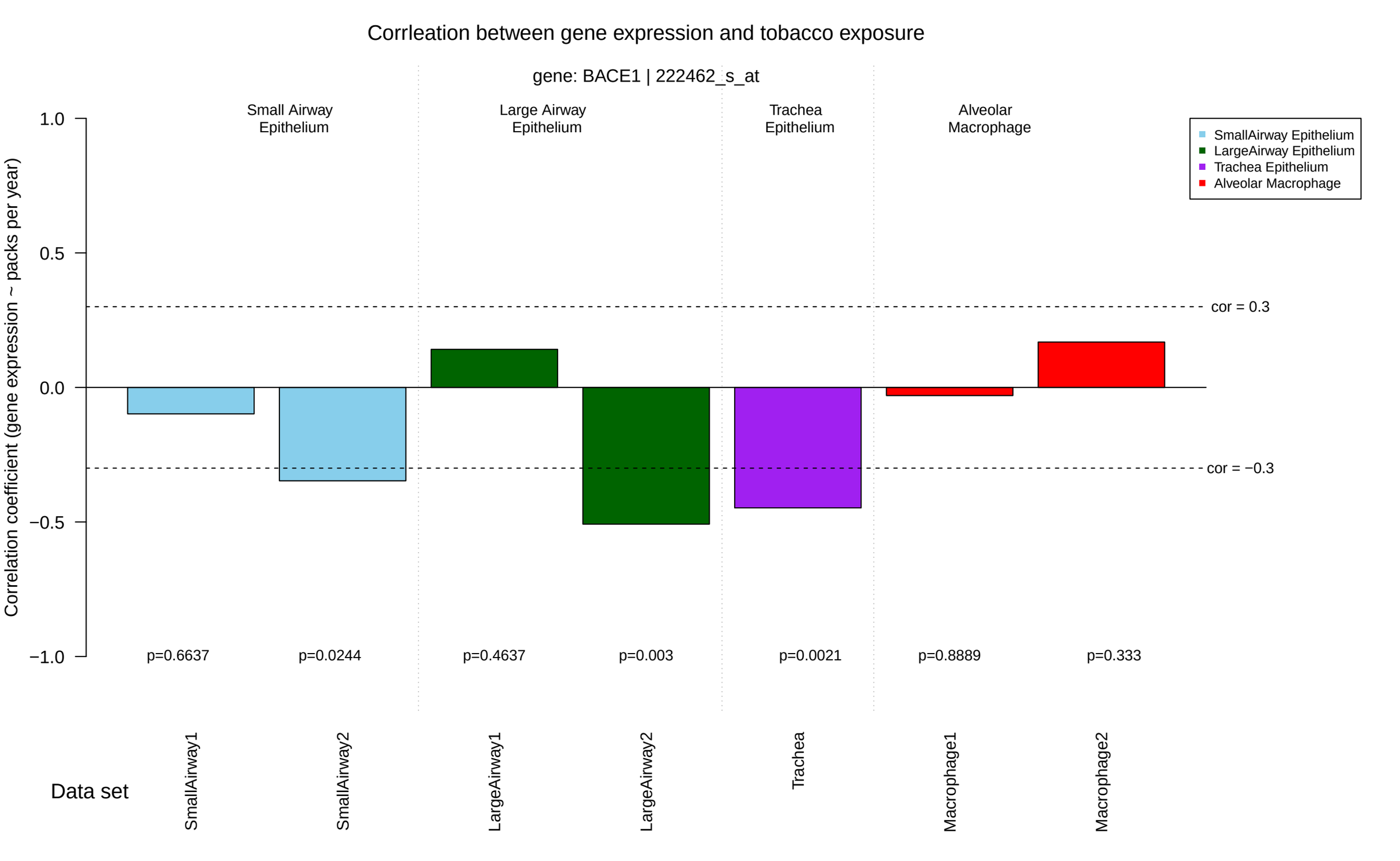

set: LargeAirway1 (p=0.8037)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0219)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4782)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2223)
BACE1 | 222462_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BACE1 | 222462_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
Correlation coefficient (gene expression ~ packs per year)
p=0.6637
p=0.0244
p=0.4637
p=0.003
p=0.0021
p=0.8889
p=0.333
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2